

PJT Report

Students :

POUILLY Léo

BREUVART Kylian

Instructor :

ROMANO Francesco

Table of contents

I)	Présentation	3
II)	Objectifs	3
III)	Création du modèle	Erreur ! Signet non défini.
IV)	Principe de convergence de maillage.....	Erreur ! Signet non défini.
V)	Choix du modèle physique	Erreur ! Signet non défini.
VI)	Convergence du maillage pour le modèle choisit.....	Erreur ! Signet non défini.
VII)	Étude approfondie de notre modèle	Erreur ! Signet non défini.

CFD on a glider wing

I) Presentation

As part of our study on the wing profile of a glider, we modeled the CLARK-Y airfoil using the CCM+ software. The choice of this profile comes from the preliminary sizing carried out in the first part of our project. The goal of this modeling is to study the behavior of our airfoil in three different flow regimes: laminar, turbulent, and inviscid.

Design specifications reminder:

$M = 500$ kg: glider mass

$V = 80$ km/h: glider flight speed

$L = 22$ m: total wingspan

$c = 1.25$ m: chord length

II) Objectives

The main objective of this lab work is to analyze the airflow around a wing profile using CCM+ in order to determine the lift and drag coefficients, which are key parameters in aerodynamics. To do so, we will study how the mesh quality influences both the results and the computation time.

The goal is to find a balance between accuracy and computational performance: a mesh that is too coarse can distort the results, whereas an overly fine mesh drastically increases computation time and resource usage. Therefore, the lab aims to test different meshing strategies, particularly by refining the mesh in critical areas around the airfoil (such as the lower surface, upper surface, and leading edge), in order to obtain reliable results while optimizing the computational co

III) Model Creation

The first step of the simulation is to import the 2D airfoil profile and ensure that the sketch is a closed contour. The sketch is then modified by closing the trailing edge of the profile. The next step is to create the simulation domain, which represents the spatial domain of our experiment.

We then move on to a crucial part of our study: meshing. The success of the entire analysis relies on this step. It is extremely important to design a mesh suited to our profile's study.

This is why we dedicate a specific section in our report to explain this process and how it enables us to obtain the desired lift and drag coefficients.

The following step consists in defining the physical characteristics of the flow — this is where we determine whether the flow is inviscid, laminar, or turbulent. Finally, we can run the simulation to extract the coefficients relevant to our study..

IV) Mesh convergence Principle

IV.1) Establishing a Coherent Mesh

As mentioned earlier, this step is crucial to our simulation. A well-designed mesh is the key to a successful simulation.

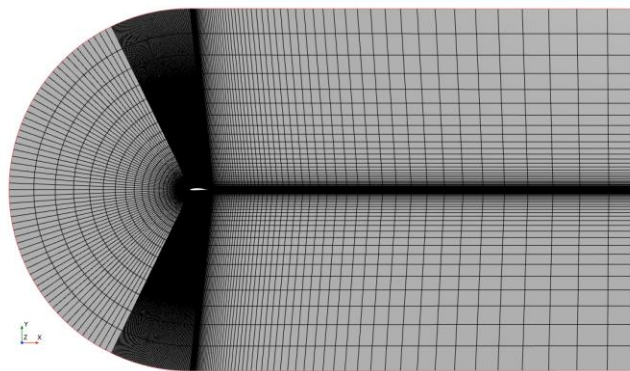


Figure 1 : Overview of the mesh

The objective is to determine the lift and drag coefficients with minimal computational cost.

A mesh with a high number of divisions is more computationally expensive than one with fewer divisions. It is therefore essential to adapt our mesh according to the different regions of the domain. Some areas will experience significant variations in the physical quantities associated with the airflow around the airfoil, while others will be much less affected.

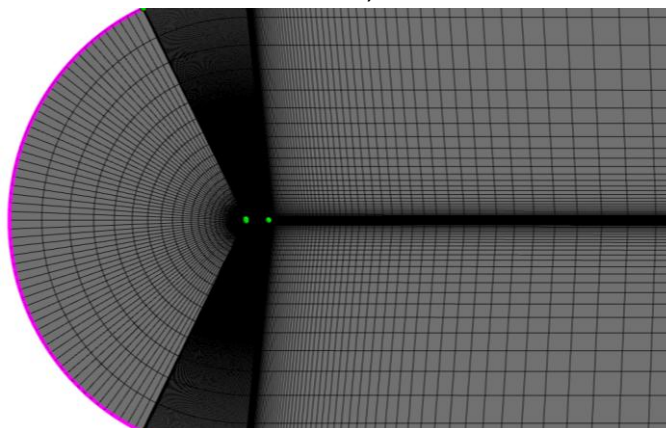


Figure 2 : Leading Edge Mesh

The area upstream of the leading edge experiences significant variations in the physical quantities related to airflow, so a fine mesh must be applied to ensure high accuracy in our simulation. In contrast, regions far from the wing profile are barely affected by these variations, so a simpler mesh is preferred to reduce computational cost. The most critical areas remain the upper and lower surfaces of the wing, where pressure variations are the largest. This shows that the key to a successful simulation is balancing accuracy and computational cost.

V) Choice of Physical Model

In the situation below, we focus on turbulent flow. Strictly speaking, a complete study should be conducted for each flow regime. Nevertheless, we will interpret each model using a default mesh previously established.

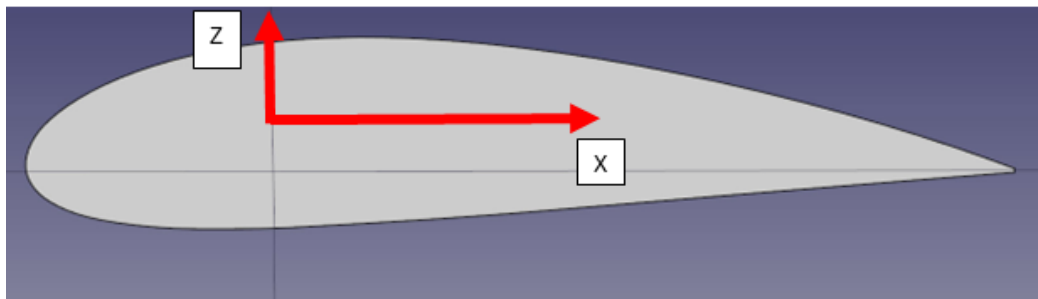


Figure 3 : Glider Wing Profile

For each flow regime, we plot the residual curves, which represent the evolution of numerical error in the equations solved during the CFD simulation convergence process. These curves thus provide a measure of local error.

V.1) Inviscid Flow

First, we perform a simulation for an inviscid fluid

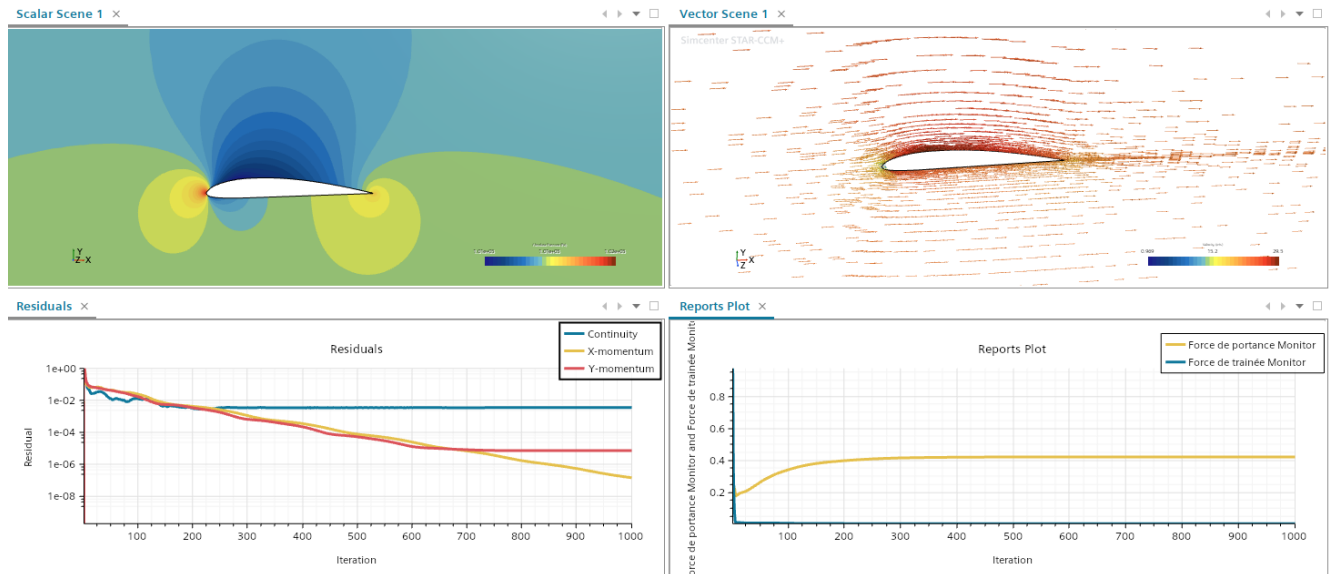


Figure 4: Residual plots, pressure gradient evolution, flow velocity evolution, and lift and drag forces evolution for inviscid flow

IV.1.a) Interprétation des courbes

Pressure Field :

High pressure (yellow/red) is observed at the front of the object, at the stagnation point where the airflow slows down significantly. On the upper surface of the wing, there is a low-pressure area (blue), typical in aerodynamics: the flow velocity increases, reducing the pressure in accordance with Bernoulli's principle. At the rear, there is an intermediate pressure zone (green), indicating a wake where pressure re-equilibrates after the passage of the object.

This pressure distribution suggests significant lift (due to the pressure difference between the upper and lower surfaces) and drag caused by high pressure at the front and low pressure in the wake.

Flow Velocity :

The velocity field is represented by colors (blue for low speed, red for high speed) and vectors indicating flow direction. High velocity (red) is observed over the top of the object, consistent with the low-pressure area in the pressure field. At the front, velocity is low (blue) as the air is compressed and slowed at the stagnation point. At the rear, vectors show a more chaotic flow with speed variations and vortices, typical of a turbulent wake.

The acceleration of the flow over the object confirms lift generation, while the turbulent wake indicates aerodynamic drag, potentially due to boundary layer separation.

Lift and Drag Forces :

Les courbes de force de portance et de traînée (Reports Plot) se stabilisent après environ 400 itérations, avec des valeurs cohérentes avec les champs observés. La force de portance est positive et constante (≈ 0.4), tandis que la force de traînée est plus faible (≈ 0.2), ce qui est attendu pour un profil aérodynamique.

Résiduals:

The "Continuity" curve drops quickly and stabilizes around $3 \cdot 10^{-3}$ from iteration 500. This indicates that mass conservation is no longer evolving but is not fully satisfied, as the residual is still high.

Ideally, a residual around 10^{-4} is needed for robust convergence. This convergence issue may arise from an under-resolved mesh or unbalanced boundary conditions.

The “X-Momentum” curve shows a steadily decreasing trend without oscillations, indicating that the X-direction momentum equation is progressively resolved, which is essential since drag is linked to this direction.

The “Y-Momentum” curve decreases and stabilizes around 10^{-4} after 700 iterations. This residual stagnation may result from lift generation causing unresolved separation, numerical instability, or improperly adjusted boundary conditions or mesh.

Iteration	Continuity	X-momentum	Y-momentum	Force de trainée	Force de portance
991	3.467481e-03	1.564110e-07	7.074581e-06	3.978323e-04	4.181365e-01
992	3.475259e-03	1.552538e-07	7.062503e-06	3.978317e-04	4.181366e-01
993	3.467481e-03	1.531764e-07	7.074564e-06	3.978312e-04	4.181366e-01
994	3.475257e-03	1.520481e-07	7.062485e-06	3.978307e-04	4.181366e-01
995	3.467480e-03	1.500222e-07	7.074546e-06	3.978302e-04	4.181366e-01

Figure 5 : Recorded Lift and Drag Forces

We now determine the lift and drag coefficients using the following relations:

$$C_z = \frac{Z}{\frac{1}{2} \cdot \rho V^2 L \cdot c}$$

$$C_x = \frac{X}{\frac{1}{2} \rho V^2 L c}$$

We obtain :

$$C_z = 5,14 \cdot 10^{-5}$$

$$C_x = 4,79 \cdot 10^{-9}$$

V.2) Écoulement laminaire

Ensuite, nous effectuons une simulation pour un écoulement laminaire.

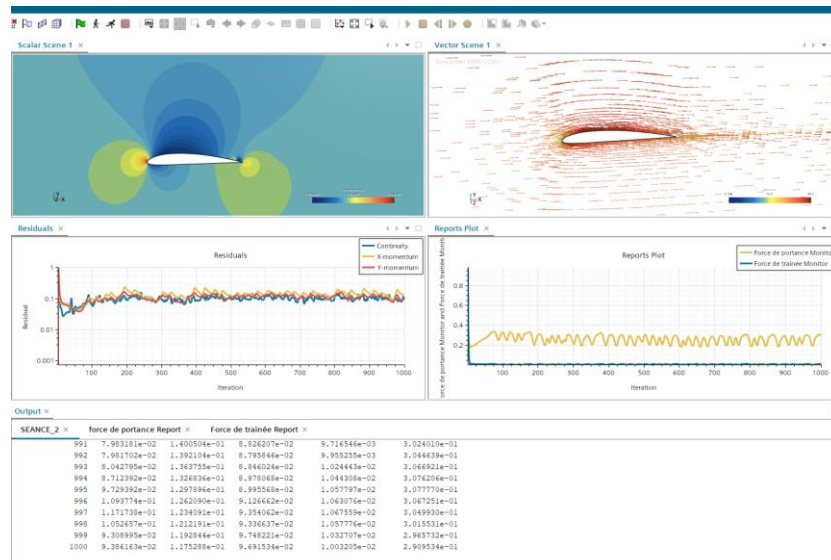


Figure 5: Residual plots, pressure gradient evolution, flow velocity evolution, and lift and drag forces evolution for inviscid flow

V.3) Turbulent Flow

As previously mentioned, we will select a model for a specific flow regime: turbulent flow. Several criteria are important for choosing an appropriate model:

- Difficulty of model convergence
- Model robustness: Ability to maintain computational performance despite changes in physical parameters
- y^+ related to y , the thickness of the cell in contact with the wall

V.3.a) Spallart-Allmaras Model

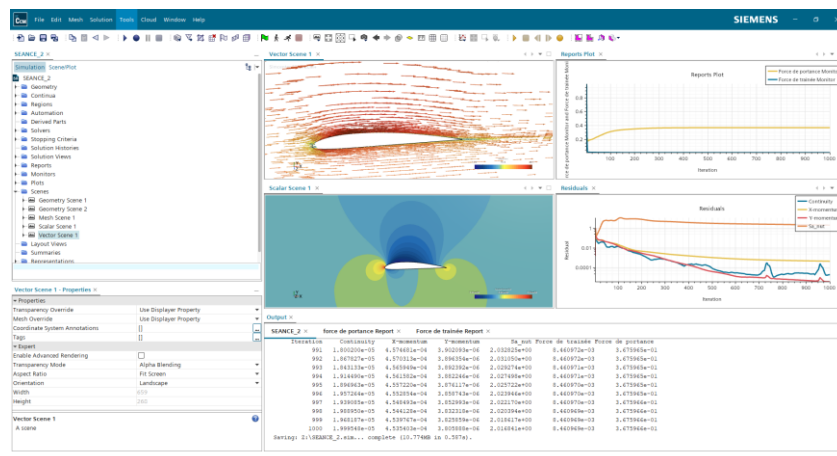


Figure 6: Residual plots, pressure gradient evolution, flow velocity evolution, and lift and drag forces evolution for turbulent flow (Spalart-Allmaras model)

V.3.b) K-Oméga Model

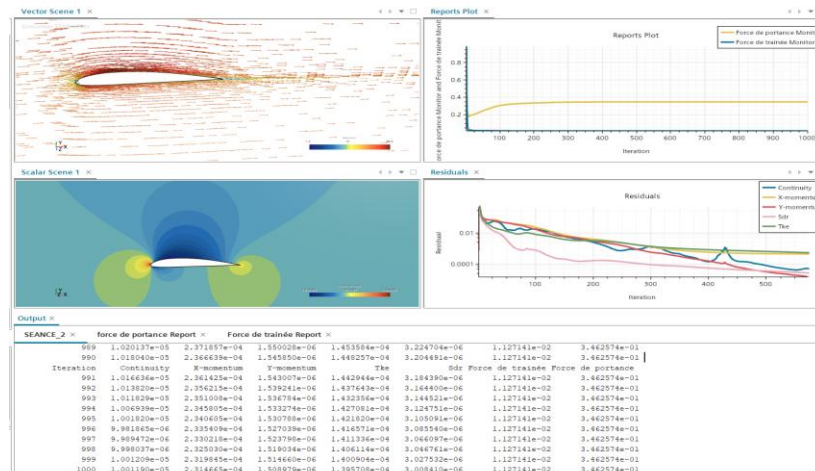


Figure 7: Residual plots, pressure gradient evolution, flow velocity evolution, and lift and drag forces evolution for turbulent flow (K-Omega model)

V.3.c) K-Epsilon Model

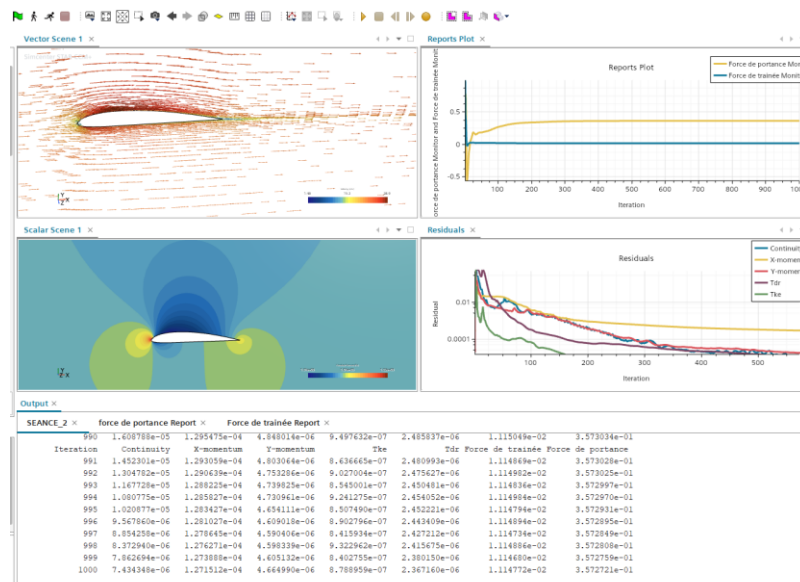


Figure 8: Residual plots, pressure gradient evolution, flow velocity evolution, and lift and drag forces evolution for turbulent flow (K-Epsilon model)

The K-Epsilon turbulence model was chosen for the simulation of the flow around the wing due to several advantages:

- **Convergence:** This model shows excellent numerical stability. Residuals decrease steadily without significant oscillations, and aerodynamic forces (lift and drag) stabilize quickly, confirming good convergence capability.

- **Robustness:** The realizable K-Epsilon model is known for its robustness against mesh and boundary condition variations. It maintains computational performance even with physical changes, making it reliable for various simulations.
- **Compatibility with y^+ :** This model is compatible with all y^+ treatments, allowing moderate meshing without excessive refinement near walls. This suits the mesh used in this study, where most y^+ values are above 30.

In comparison, other models such as K-Omega are more sensitive to mesh and require low y^+ treatment, making implementation more complex and less stable in this context. The K-Epsilon model therefore offers a balanced solution between accuracy, stability, and ease of use.

VI) Mesh Convergence for the chosen model

We now test increasing mesh precision to determine if our values are reliable. The goal is to select the least computationally expensive mesh that provides accurate results.

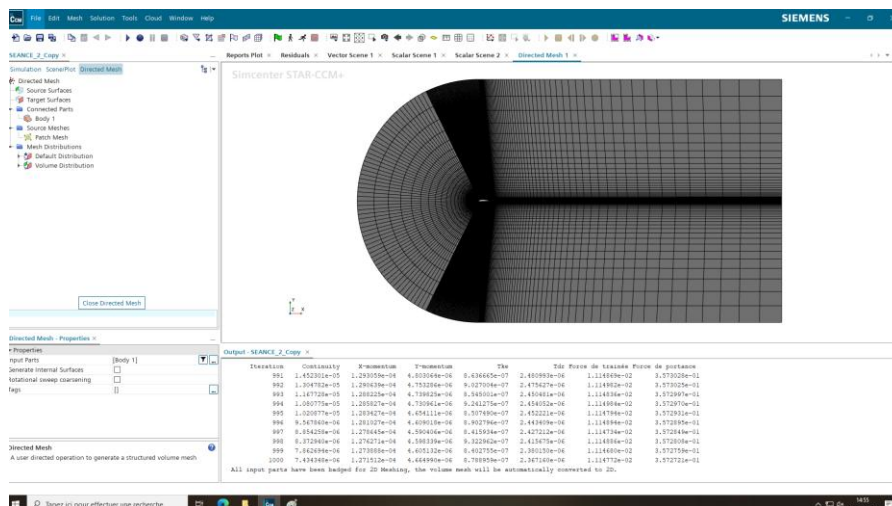


Figure 9 : Initial mesh for the K-Epsilon model

Initially, a mesh is chosen, and lift and drag forces are recorded. The mesh is then modified to study convergence of these values. Mesh selection follows the criteria explained in Section IV of this report.

VI.1) Evaluation of Lift Coefficient as a Function of Mesh

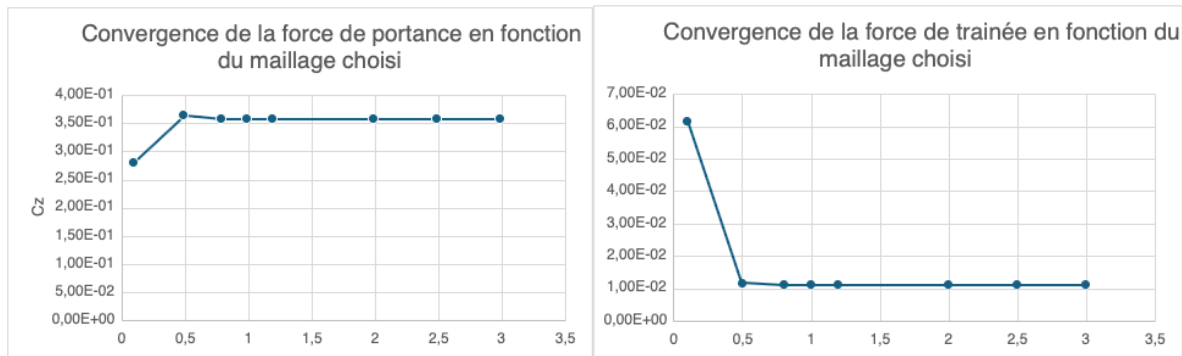


Figure 10: Evolution of lift and drag forces as a function of mesh

The selected mesh corresponds to 80% of the initial mesh size, offering a good compromise between result accuracy and computational cost, while ensuring numerical stability and simulation convergence.

VII) In-depth study of our model

VII.1) Influence of Angle of Attack on C_z

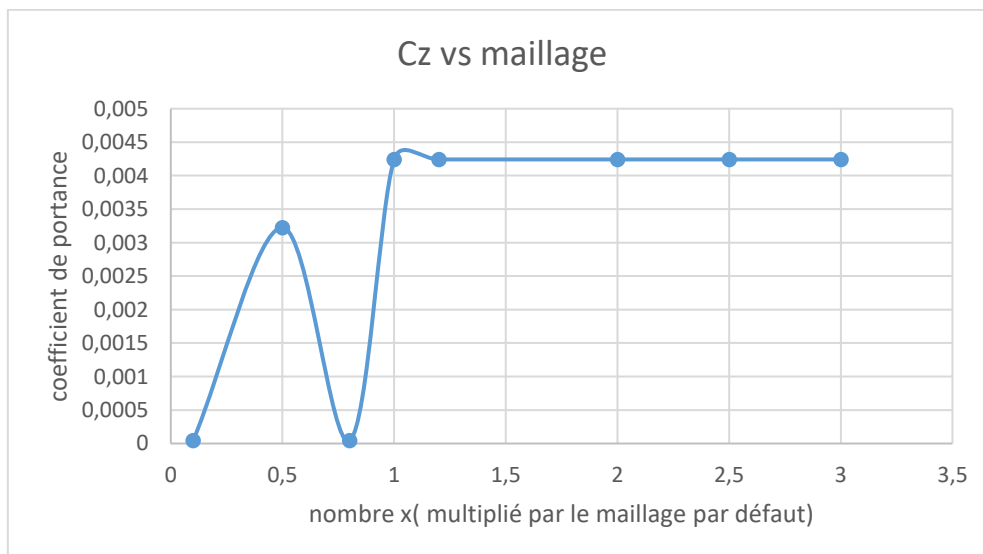


Figure 11: Evolution of lift coefficient as a function of mesh

Up to $x \approx 1$, C_z varies significantly, showing numerical oscillations caused by a too coarse mesh or poor resolution of the pressure field around the profile. The lift coefficient suggests increasing the mesh refinement to 1.2 instead of 0.8, which is still reasonably inexpensive, achieving proper mesh convergence.

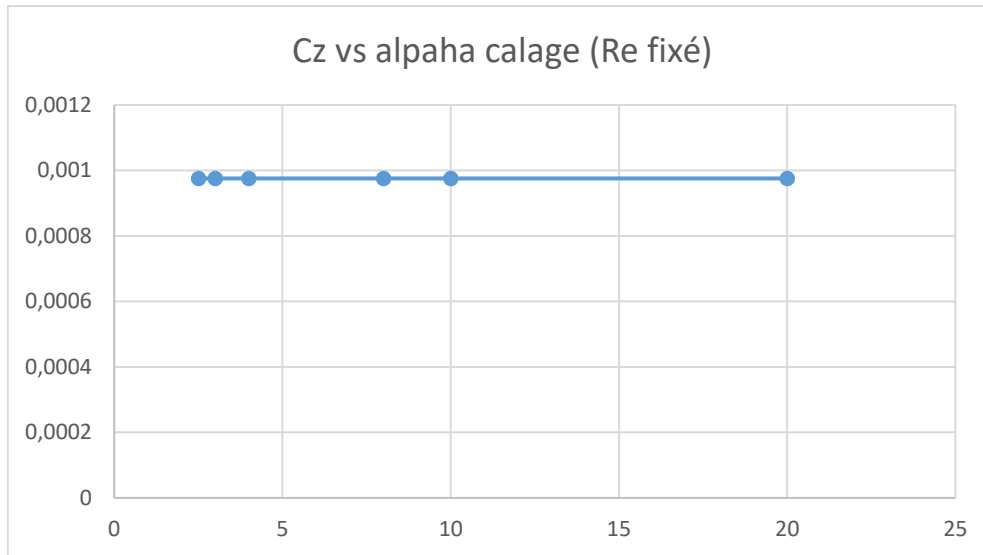


Figure 12: Evolution of lift coefficient as a function of angle of attack

In this curve, the lift coefficient C_z remains almost constant despite increasing the angle of attack. This result is surprising, as theoretically, C_z should increase with the angle of attack until stall. This suggests a modeling or configuration issue in the simulation. It is possible that the angle of attack is not correctly applied to the geometry or that the forces are misprojected, distorting the lift calculation. However, care was taken to define the lift force in the moving reference frame.

VII.2) Influence of Reynolds Number on C_z

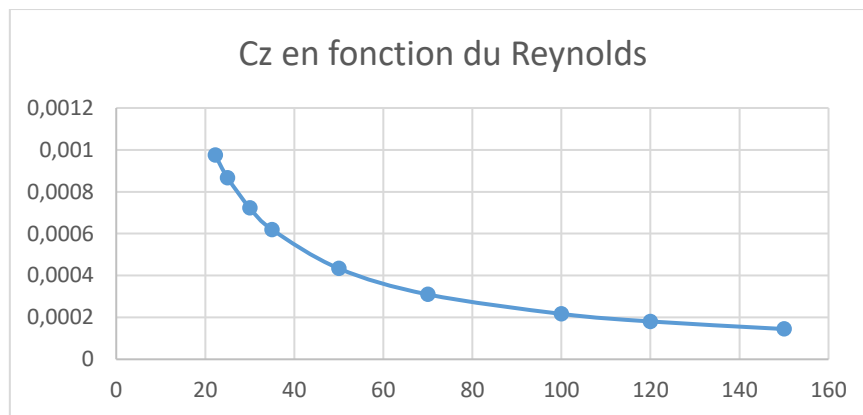


Figure 13: Evolution of the lift coefficient as a function of the Reynolds number

Here, we observe that the lift coefficient C_z decreases as the Reynolds number increases, which is the opposite of the expected aerodynamic behavior. This can be explained by the use of the $k-\varepsilon$ model, which is unreliable at low Reynolds numbers because it overestimates turbulent losses. Furthermore, a mesh that is not well adapted to the different Reynolds values may amplify this error. This behavior highlights the need for a turbulence model better suited to transitional or low-Reynolds-number flows.

In this turbulent flow study using the $k-\varepsilon$ model, we analyzed the influence of the mesh, the angle of attack, and the Reynolds number on the lift coefficient C_z . The results show good mesh convergence, validating the numerical stability of the simulation. However, the variations of C_z with the angle of attack and Reynolds number reveal physical limitations of

the model used. The abnormally constant behavior of C_z with respect to the angle of attack suggests either an incorrect application of the angle or an issue with the force projection. Moreover, the decrease of C_z with increasing Reynolds number, contrary to physical reality, underlines the weaknesses of the k - ε model at low Reynolds numbers. These results emphasize the importance of choosing an appropriate turbulence model and ensuring rigorous parameterization to achieve reliable aerodynamic predictions.